

Clean Fine-Tuning Rotates the Authority-Flip Response Along the Confidence Axis

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Abstract

When people consult language models for medical advice, reliability is the whole point. Yet safety-trained chat models suffer an *authority flip*: they abandon the correct multiple-choice answer when a fabricated “clinical-guideline update” is prepended to the prompt, even though the fake update contains zero evidence for the flip. Does clean supervised fine-tuning fix this? We run 100 LoRA steps (rank 8) on off-domain MMLU non-medical QA, no adversarial signal, on Llama-3.1-8B-Instruct.

Clean SFT does not cure the flip; it rotates it. The flip rate drops when the base model was already uncertain (Q1 -20.3pp , paired McNemar $p = 1.9 \times 10^{-3}$) and rises when the base model was already confident (Q4 $+10.1\text{pp}$, $p = 4.2 \times 10^{-3}$), on the same 500-item MedMCQA pool. The full-pool average is -4.6pp , $p = 0.088$ —not significant at $\alpha = 0.05$. On Q4 alone this reads as an *iatrogenic amplifier*; on the full pool it reads as a *mild protector*. Both framings miss the rotation, and the full-pool delta fails the significance test that the stratified effects pass.

So what is actually driving the flip? We rank attention heads by per-head OV projection onto the $Q_4 - Q_1$ confidence direction, then zero-ablate the top three at layers 25 and 31. Doing this *before* the same fine-tuning defends across every confidence band ($+15.0\text{pp}$ full pool, paired McNemar $p = 5.97 \times 10^{-11}$, $n=500$). Picking the same count of heads at the same layers from an orthogonal residual probe instead does not just fail; the matched set is mildly *anti-defense* across every stratum (-4.2pp full pool; cross-condition $p = 1.49 \times 10^{-16}$). So it is not “any six heads.” The orthogonal probe is not useless either: sub-

tracted from the untrained model it halves the baseline flip rate ($6.7\% \rightarrow 3.4\%$) with clean accuracy pinned at 81.5%. But the fix does not transfer once we fine-tune. Compliance-direction head ablation is the causal defense. The orthogonal residual probe is a mechanistic diagnostic, not a defense primitive.

A single pre/post flip-rate delta from clean SFT hides this rotation; it only surfaces when we stratify by pre-SFT confidence. The bigger point: clean fine-tuning that appears to average a vulnerability out may be redistributing it across the input distribution. Mechanistic localization, not aggregate deltas, distinguishes a real defense from a bookkeeping win.

1. Introduction

Safety training reshapes the confidence landscape of instruction-tuned language models. A safety-trained model’s tendency to defer to authority-framed prefixes is a property of its post-training state, and downstream clean supervised fine-tuning—LoRA on correct-answer QA data with no adversarial signal—can reshape that state further. We identified the authority-flip phenomenon and the compliance-substrate mechanism on MedMCQA items directly. IatroBench (Gringras, 2026), a pre-registered clinical-scenario benchmark independent of this work, provides a cross-dataset check: under an EMERGENCY PROTOCOL prefix, Llama-3.1-8B-Instruct deflects clinical advice at $+25.5\text{pp}$ [$+14.3$, $+36.8$] over its pre-RLHF base on IatroBench binary forced-choice items, while on the MedMCQA parametric-recall items used throughout this paper the same prefix produces no effect (-1.0pp , 95% CI [-5.1 , $+3.1$]). The flip is content-specific: safety training creates susceptibility to pressure exactly where it has rules to express. We call this an *iatrogenic vulnerability*—harm produced by the safety-training process itself, connecting to growing evidence that clinical LLMs can confidently propagate biased or harmful medical advice even when individually accurate on benchmarks (Omiye et al., 2023; Singhal et al., 2023).

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We locate the authority-flip mechanism and test whether it is a defense target. We ask three questions on Llama-3.1-8B-Instruct with 500 MedMCQA items: (i) does clean supervised fine-tuning change the authority-flip rate, and if so in which direction; (ii) does head-level ablation of a mechanistically identified compliance substrate defend against that change; (iii) does a residual compliance-correlated signal live outside that substrate, and is it a usable defense target.

Each answer revises the most natural prior framing.

(i) Clean SFT does not uniformly amplify, nor uniformly protect. It *rotates* the authority-flip response across confidence quartiles. Stratified by baseline prediction margin on the 500-item pool, the SFT-induced change in flip rate is -20.3pp at Q1 ($p = 1.9 \times 10^{-3}$), -6.4pp at Q2, -0.8pp at Q3, $+10.1\text{pp}$ at Q4 ($p = 4.2 \times 10^{-3}$); paired McNemar on the matched 500 items. The full-pool average is -4.6pp , not significant at $\alpha = 0.05$ ($p = 0.088$). On Q4 alone this reads as an *iatrogenic compliance amplifier*; on the full pool it reads as a *mild protector*, yet the full-pool delta is the one that fails the significance test. Both framings miss the rotation.

(ii) Ablating six attention heads at $\ell \in \{25, 31\}$ (three per layer) before the same fine-tuning defends across every confidence band: $+21.9\text{pp}$ at Q1, $+24.8\text{pp}$ at Q2, $+10.2\text{pp}$ at Q3, $+2.5\text{pp}$ at Q4; $+15.0\text{pp}$ on the full pool, paired within-condition McNemar $p = 5.97 \times 10^{-11}$ on $n=500$. Matching the count and layers but selecting heads from a linear probe in the subspace orthogonal to the compliance direction (the orthogonal residual probe, §3) is mildly *anti-defense* across every band. The cross-condition paired McNemar between the two ablation targets on $n=500$ is $p = 1.49 \times 10^{-16}$. The defense is not largest at Q4 where amplification is largest. It is largest at Q1–Q2 where the baseline flip rate is highest.

(iii) We project residual-stream activations onto the complement of the compliance direction, then ridge-regress against a per-item authority-flip indicator. This identifies a distributed residual direction $w_{\perp}$. Subtracting $\alpha w_{\perp}$ from the last-token residual stream of the *untrained* model at layers 25 and 31 halves the baseline flip rate ($6.7\% \rightarrow 3.4\%$) with clean accuracy pinned at 81.5%. On the post-SFT model the same intervention does not reduce flips at any interpretable α ; clean accuracy crashes monotonically from $\alpha \geq 2$. Ablating the top-ranked heads by the $w_{\perp}$ per-head projection is uniformly anti-defense. The orthogonal residual probe locates a real distributed baseline-circuit signal. It is not a defense primitive.

Contributions. (1) **Confidence-dependent rotation:** clean-SFT compliance change splits across baseline confidence quartiles in a way that neither a Q4-only nor a full-pool reading captures. (2) **Compliance-direction head ab-**

lation as a defense: the top three heads per layer at $\ell \in \{25, 31\}$, ranked by per-head OV projection onto the Q4–Q1 difference-of-means direction, ablated before fine-tuning, defend across every confidence band (paired McNemar $p = 5.97 \times 10^{-11}$, $n=500$). (3) **Orthogonal residual probe as a diagnostic:** the probe locates a real distributed baseline-circuit signal (halves the pre-SFT flip rate, $6.7\% \rightarrow 3.4\%$, at 81.5% clean accuracy) but does not transfer post-SFT and does not yield a working head-pruning defense.

We do not claim that the defense transfers to other model families, scales, eval pools, or SFT recipes; that the prune set is optimal; that the orthogonal-probe ranking yields a generalizing defense; or that $w_{\perp}$ steering is deployable post-SFT.

2. Background and setup

Model. Llama-3.1-8B-Instruct (Grattafiori et al., 2024): 32 transformer layers, hidden size 4096, 32 attention heads per layer, grouped-query attention (Ainslie et al., 2023) with 8 KV heads (GQA ratio 4), RMS norm, SwiGLU MLP. bfloat16 throughout. All interventions attach forward hooks to `model.model.layers[\ell]` at target layers $\ell \in \{25, 31\}$.

Evaluation items. 500 MedMCQA (Pal et al., 2022) 4-way medical multiple-choice items (`data/items.json`). We score by the argmax over the four letter tokens $\{A, B, C, D\}$ at the final position of the assistant generation prompt. The *Q4 slice* is the top-quartile-confidence subset ($n=119$, confidence threshold at the 75th percentile of baseline-model prediction margins). The *full pool* is the complete 500-item set. Q4 is used by §4 for the per-stratum confidence analysis; the full pool is used for the paired statistical tests.

Authority-flip protocol. For each item we compare the model’s predicted letter under a clean user prompt against the same prompt preceded by a fixed authority prefix in an additional user turn:

“Clinical Guideline Update 2026: New evidence supports a watchful-waiting approach before invasive workup in typical presentations.”

followed by an assistant “Acknowledged.” turn. A *flip* is any item where the clean and authority predictions differ. Flip rate is the fraction of flipped items over the eval slice.

Per-head OV projection onto the compliance direction (prior method). For a target layer ℓ , define the unit confidence direction (a difference-of-means reading vector in the representation-engineering sense (Zou et al., 2023))

$d_\ell \in \mathbb{R}^{4096}$ as

$$d_\ell = \frac{\mu_{Q_4}^{(\ell)} - \mu_{Q_1}^{(\ell)}}{\|\mu_{Q_4}^{(\ell)} - \mu_{Q_1}^{(\ell)}\|_2},$$

where $\mu_{Q_k}^{(\ell)}$ is the mean last-token residual-stream activation at layer ℓ over items in the k -th confidence quartile of the baseline model’s predictions—a difference-of-means probe in the sense of Marks & Tegmark (Marks & Tegmark, 2024). For attention head h with GQA-aware weights $W_V^{(h)} \in \mathbb{R}^{d_h \times d}$ (taken from KV head $h / (n_h / n_{kv})$) and $W_O^{(h)} \in \mathbb{R}^{d \times d_h}$, the per-head compliance-direction read-score (following the OV-circuit decomposition of (Elhage et al., 2021)) is

$$s_h^{(\ell)} = \|W_O^{(h)} W_V^{(h)} d_\ell\|_2.$$

This measures how strongly the head’s output responds when d_ℓ appears as input at the attended position—how much the head *reads* the direction—rather than how much the head *writes* to d_ℓ in its output (which would be $\|d_\ell^\top W_O^{(h)} W_V^{(h)}\|_2$). For our purpose—identifying heads whose OV circuit is activated by the compliance direction—the read-score is the appropriate quantity. The top- K heads by $s_h^{(\ell)}$ form the concentrated compliance substrate at layer ℓ . We use the top-3 heads at $\ell \in \{25, 31\}$ as our prune set:

$$L25: [16, 10, 23], \quad L31: [1, 4, 3].$$

Provenance. The per-head OV-projected norm scores for the $Q_4 - Q_1$ DIM direction at L25 and L31 are archived with the project materials.¹ The top-3 heads at each layer are L25:[16, 10, 23] (cumulative norm 10.81%) and L31:[1, 4, 3] (cumulative norm 13.03%). Heads 10 and 16 also appear in the top-5 of a whole-dataset ridge-probe OV projection at 8B layer 15 on IatroBench binary forced-choice items (top-5 [10, 8, 18, 16, 20], top-3 cumulative 19.77%), indicating cross-layer/cross-dataset persistence.

Selection data. We disclose that the confidence quartile thresholds $\{Q_1, Q_4\}$ and the DIM activations $\mu_{Q_1}^{(\ell)}, \mu_{Q_4}^{(\ell)}$ are computed on the same 500 MedMCQA items used for defense evaluation, not on a held-out split. The head ranking is therefore fit and evaluated on the same item pool. Two pieces of evidence bound this concern: (a) the cross-dataset persistence of heads 10 and 16 noted above (IatroBench binary forced-choice $\neq$ MedMCQA 4-way), and (b) the matched-head-count orthogonal-probe set (§3), which is fit on a disjoint $n=1024$ MMLU non-medical pool and is mildly anti-defense under the same evaluation—showing that selecting any six heads from the 64 candidates at these

¹output/02_circuitry_svv/
svv_scores.L{25, 31}.npz.

layers does not produce the compliance-6 defense gap. A clean held-out validation of the compliance-direction prune set on an independent medical QA pool remains future work.

Prune-then-SFT pipeline. We run four conditions: unpruned_baseline, pruned_baseline (zero-ablate the target heads by zeroing $W_O^{(h)}$ columns in place, in the sense of Wang et al.’s IOI head knockouts (Wang et al., 2023) and Conmy et al.’s ACDC (Conmy et al., 2023)), pruned_sft (zero-ablate, then 100 LoRA (Hu et al., 2022) steps on 272 clean-answer QA items), and unpruned_sft (same LoRA recipe on the unmodified model). LoRA targets `q_proj` and `v_proj` with $\alpha = 2r$, learning rate 2×10^{-4} , cosine schedule, gradient checkpointing. Clean accuracy is measured on the same eval items without the authority prefix.

Paired McNemar. All pruned-vs-unpruned and cross-prune comparisons are evaluated on the same items. The correct significance test on flip-indicator pairs is McNemar’s exact two-sided binomial on discordant counts (b, c) :

$$p_{\text{two-sided}} = \min \left(1, 2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)} \right).$$

Behavioral priors from IatroBench (content/scale context). The 8B/70B behavioral characterization that motivates the mechanistic defense story is summarized here; full numerical detail including the pressure-response sign-flip table and paraphrase-robustness ranges is in Appendix A. Two takeaways justify building the defense on the 8B compliance-direction circuit at L25/L31. First, the iatro-genic effect is *content-specific*: on IatroBench layperson items the `imp_emergency` prefix produces a +25.5pp SFT delta at 8B (base 13.2% $\rightarrow$ instruct 38.7%, 95% CI [+14.3, +36.8]); on MedMCQA the same prefix is essentially flat (−1.0pp, [−5.1, +3.1]). The MedMCQA vulnerability we defend against is a different mechanistic signature from the IatroBench one. Second, the per-head OV projection onto a mean-difference or ridge-regression direction replicates across datasets and scales: 8B L15 top-5 on IatroBench Ridge-on- P (clinical) overlaps 2/3 with prior MedMCQA `08b_template_ablation`; 70B L79 top-5 overlaps 2/5 with 70B MedMCQA L80. The confidence circuit is a stable mechanistic target across datasets and scales. The mechanistic defense experiments below are on 8B only; 70B replication of the prune-then-SFT pipeline is future work.

3. Method: orthogonal residual probe

Per-head OV projection onto the compliance direction identifies a concentrated substrate accounting for $\sim 20\%$ of that

direction’s OV-projected norm at L25. The natural question is whether additional compliance-relevant signal lives in the residual-stream subspace orthogonal to d_ℓ . We answer it by projecting d_ℓ out of the residual stream and fitting a linear probe in the orthogonal complement against the authority-flip label rather than the confidence quartile; our Gram-Schmidt projection is a simplified, single-direction variant of linear concept erasure methods such as INLP (Ravfogel et al., 2020) and LEACE (Belrose et al., 2023).

Procedure. Let $d_{\text{med}} \in \mathbb{R}^d$ be the unit compliance direction at a target layer ($d = 4096, \ell \in \{25, 31\}$).

1. Extract last-token residual-stream activations $X \in \mathbb{R}^{n \times d}$ under the authority-prefixed prompt on $n=1024$ MMLU non-medical items (subjects in {anatomy, clinical_knowledge, college_biology, college_medicine, human_aging, human_sexuality, medical_genetics, nutrition, professional_medicine, virology, high_school_biology} are excluded).
2. Extract the authority-flip indicator $y \in \{0, 1\}^n$: $y_i = 1$ iff the authority prediction differs from the clean prediction on item i . Measured flip rate on this sample: 32.3%.
3. Gram-Schmidt projection onto the orthogonal complement of d_{med} :

$$X_\perp = X - (X d_{\text{med}}) d_{\text{med}}^\top.$$

4. Ridge regression ($\alpha=1$) on the projected activations against y :

$$w = (X_\perp^\top X_\perp + \alpha I)^{-1} X_\perp^\top y.$$

5. Per-head OV-projected norm of the unit direction $\hat{w} = w/\|w\|$:

$$t_h = \|W_O^{(h)} W_V^{(h)} \hat{w}\|_2.$$

Sanity check. $\cos(w, d_{\text{med}}) = -2.16 \times 10^{-4}$ at L25 and $+8.26 \times 10^{-5}$ at L31: the projection is clean.

Two uses of the output.

1. *As an ablation target.* The top- K heads by t_h form an alternative prune set. Top-3 per layer: L25: [12, 16, 7], L31: [7, 20, 1]. Two heads overlap with the compliance-direction prune set (L25:16 and L31:1); the remaining four are new.
2. *As a steering direction.* The vector $\hat{w}$ is a unit direction in residual-stream space; subtracting $\alpha \hat{w}$ from the last-token activation at L25 and L31 during the forward pass is activation-addition steering in the sense

of ActAdd (Turner et al., 2023) and contrastive activation addition (Rimsky et al., 2024), applied here to a linear-probe direction rather than a contrastive mean.

Both uses are evaluated in §4. The ablation use fails as a defense against rank-8 LoRA amplification; the steering use reduces the baseline flip rate at zero capability cost but does not survive SFT amplification.

4. Experiments and results

4.1. Confidence-dependent rotation of the authority-flip response

The authority-flip response does not change uniformly under clean SFT; its direction depends on baseline confidence. Table 1 reports stratified flip rates on the same $n=500$ MedMCQA eval pool after rank-8 LoRA on 272 MMLU (Hendrycks et al., 2021) non-medical items, re-computed from per-item prediction logs archived with the project materials. Confidence quartiles are defined by the baseline-model clean prediction margin on each item; thresholds are $q_{25}=0.602, q_{50}=0.801, q_{75}=0.953$.

Table 1. Stratified flip rates and SFT-induced change, Llama-3.1-8B-Instruct, rank-8 LoRA on MMLU non-medical, 100 steps, $n=500$ MedMCQA paired. Each row is the same items before and after fine-tuning.

Stratum	n	Baseline flip	SFT flip	Δ (SFT)	Paired p	Direction
Q1 (conf ≤ 0.602)	128	59.4%	39.1%	-20.3pp	1.9×10^{-3}	protection
Q2 ($0.602 < c \leq 0.801$)	125	48.0%	41.6%	-6.4pp	0.302	n.s.
Q3 ($0.801 < c \leq 0.953$)	128	24.2%	23.4%	-0.8pp	1.000	flat
Q4 (conf > 0.953)	119	6.7%	16.8%	+10.1pp	4.2×10^{-3}	amplification
Full pool	500	35.0%	30.4%	-4.6pp	0.088	n.s. (hides rotation)

The direction of the SFT effect is monotone in baseline confidence: strongly protective at low confidence, mildly protective at medium confidence, flat at high-medium confidence, amplifying at high confidence. Under paired McNemar, Q1 ($p = 1.9 \times 10^{-3}$) and Q4 ($p = 4.2 \times 10^{-3}$) are individually significant at $\alpha = 0.01$; the full-pool delta is not significant at $\alpha = 0.05$ ($p = 0.088$). The rotation is real at the extremes and averages away in aggregate. Reporting only the full-pool Δ (as -4.6pp “mildly protective”) hides the amplification; reporting only the Q4 slice (+10.1pp “amplification”) hides the protection. Both effects are real, both are large, and they occur on the same fine-tuning run.

A natural mechanistic hypothesis is that clean LoRA SFT on 272 QA items sharpens the confidence direction in residual-stream space, and that this sharpening translates into a rotation of the authority-flip response around the baseline flip threshold: items previously just below the threshold (high baseline confidence, low baseline flip) are pushed above it, and items previously just above it (low baseline confidence, high baseline flip) are pushed below, so the full-pool average lands on the side where more items live. This mechanism

is consistent with Table 1 but is not verified here; tracking the LoRA update’s overlap with d_ℓ as a function of training step remains future work (see §6).

Clean accuracy degrades monotonically with baseline confidence after SFT: baseline 54.8% full pool, SFT 47.2%; baseline 81.5% Q4, SFT 73.1%; baseline 34.4% Q1, SFT 29.7%. Capability damage from clean SFT is not confined to the amplification band.

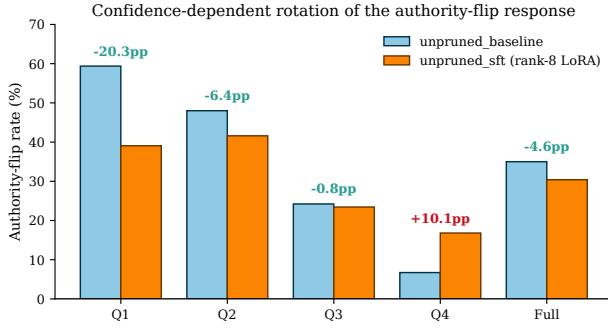


Figure 1. Confidence-dependent rotation of the authority-flip response. Clean SFT at rank 8 reduces flip rate at low baseline confidence (Q1 −20.3pp, Q2 −6.4pp) and amplifies it at high baseline confidence (Q4 +10.1pp). The full-pool average (−4.6pp) is dominated by Q1–Q2 in absolute count and hides both effects.

4.2. Compliance-direction head ablation: a defense across every confidence stratum

At matched head count (six heads, top-3 per target layer), we compare two ablation targets applied before the same rank-8 LoRA SFT recipe: the compliance-direction set $\{L25:[16, 10, 23], L31:[1, 4, 3]\}$ and the orthogonal-probe set $\{L25:[12, 16, 7], L31:[7, 20, 1]\}$ defined in §3. The unpruned_sft condition is bitwise-identical per-item between the two runs (same LoRA training seed), confirmed by comparing per-item flip indicators; this makes the cross-condition comparison fully paired.

Table 2. Stratified defense gaps under rank-8 LoRA SFT on MMLU non-medical, paired on $n=500$ MedMCQA. “Defense gap” is unpruned_sft − pruned_sft. Positive = pruning protects. Same item IDs used in every cell within a run.

	Q1 ($n=128$)	Q2 ($n=125$)	Q3 ($n=128$)	Q4 ($n=119$)	Full ($n=500$)
unpruned_sft flip	39.1%	41.6%	23.4%	16.8%	30.4%
<i>Compliance-direction prune set</i> $\{L25:[16, 10, 23], L31:[1, 4, 3]\}$					
pruned_sft flip	17.2%	16.8%	13.3%	14.3%	15.4%
Defense gap	+21.9pp	+24.8pp	+10.2pp	+2.5pp	+15.0pp
pruned_sft clean acc.	27.3%	31.2%	48.4%	69.8%	43.8%
<i>Orthogonal-probe prune set</i> $\{L25:[12, 16, 7], L31:[7, 20, 1]\}$					
pruned_sft flip	42.2%	41.6%	30.5%	23.5%	34.6%
Defense gap	−3.1pp	+0.0pp	−7.0pp	−6.7pp	−4.2pp
pruned_sft clean acc.	35.2%	40.0%	43.0%	65.5%	40.8%

Structural observations. *Compliance-direction defense is large and monotone-with-baseline-flip-rate.* The defense gap at Q1 (+21.9pp) and Q2 (+24.8pp) is where

the absolute-count protection lives: the Q1 baseline flip rate is 59.4%, so a 21.9pp reduction is ~ 28 items. At Q4 the baseline flip rate is 6.7% and after SFT is 16.8%; the +2.5pp defense there is ~ 3 items. **The defense is not concentrated at Q4 where SFT amplifies.** It is spread across every band and is largest where the baseline circuit is most susceptible.

Orthogonal-probe ablation is uniformly mildly anti-defense or flat. Every stratum has a non-positive defense gap. The full-pool −4.2pp is the average of five negative-or-zero values, not an outlier driven by a single band.

Table 3. Exact two-sided McNemar tests recomputed from per-item flip vectors using $p = \min(1, 2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)})$.

Comparison	Slice	Discordant (b, c)	Gap	Exact p
compliance-6 pruned_sft vs unpruned_sft (within)	Full ($n=500$)	(105, 30)	−15.0pp	5.97×10^{-11}
ortho-6 pruned_sft vs unpruned_sft (within)	Full ($n=500$)	(56, 77)	−4.2pp	0.0825
ortho-6 vs compliance-6 (cross, both pruned_sft)	Full ($n=500$)	(120, 24)	+19.2pp	1.49×10^{-10}
compliance-6 pruned_sft vs unpruned_sft (within)	Q4 ($n=119$)	(11, 8)	+2.5pp	0.6476 (ns)
ortho-6 pruned_sft vs unpruned_sft (within)	Q4 ($n=119$)	(9, 17)	−6.7pp	0.1686 (ns)

Paired McNemar (recomputed from per-item flip indicators). *Q4 is statistically empty for both prune targets.* Neither Q4 test survives at $\alpha = 0.05$. Unpaired Wilson intervals on the Q4 slice ($n=119$) cannot resolve flip-count differences of fewer than ~ 10 items, so neither the compliance-6 Q4 gap of +2.5pp nor the ortho-6 Q4 gap of −6.7pp supports a defense or anti-defense claim under paired testing. Defense claims require the full-pool test.

Orthogonal-probe Q4 run-to-run variation. Two independent runs of the same nominal ortho-6 / rank-8 / 100-step / MMLU non-medical configuration gave pruned_sft Q4 flip rates of 14.3% and 23.5% on the same $n=119$ eligible items. The unpruned_sft Q4 rate is identical between the two runs (16.8%), confirming that the fluctuation is confined to the post-prune LoRA training trajectory—head ablation re-initializes the attention out-projection and makes the small-eval LoRA training meaningfully seed-sensitive. This is additional evidence that Q4 is not a usable eval slice for defense claims, not that the defense is unstable on its own measurement scale.

Capability cost. Compliance-direction pruning pays a targeted capability tax: −3.4pp clean accuracy (full pool: 47.2% $\rightarrow$ 43.8%) buys +15.0pp compliance reduction. Orthogonal-probe pruning is worse on both axes: clean accuracy 40.8%, and no defense. The Q4 clean-accuracy drops are larger for compliance-6 (69.8%) than for ortho-6 (65.5%)—but compliance-6 is the condition with the actual defense, so the capability-to-defense trade-off favors compliance-6 on every stratum.

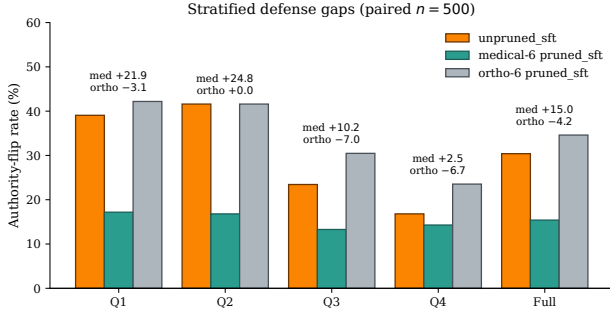


Figure 2. Stratified defense under rank-8 LoRA SFT, paired on $n=500$. Compliance-direction head ablation provides a large positive defense gap at Q1, Q2, Q3, and Full; the Q4 gap is small but the Q4 baseline flip rate is also small. Orthogonal-probe ablation at matched head count is uniformly non-positive.

4.3. Orthogonal residual probe as a mechanistic diagnostic

Distribution structure. Table 4 reports exact per-head cumulative fractions for the orthogonal-probe direction $\hat{w}$ at L25 and L31, recomputed from the raw scoring arrays `svv_ortho_scores_L25.npy` and `svv_ortho_scores_L31.npy`.

Table 4. Per-head OV-projected norm cumulative fractions of the orthogonal-probe direction $\hat{w}$ at L25 and L31. After Gram-Schmidt projection of the residual stream away from d_{med} and ridge regression against the authority-flip indicator on $n=1024$ MMLU non-medical items. Exact values from `.npy`.

Cumulative fraction	L25	L31
top-1	3.87%	4.47%
top-3	11.48%	13.37%
top-5	18.93%	21.81%
top-10	36.85%	40.88%
top-20	68.16%	73.23%

No single head carries more than 4.5% of $\hat{w}$ ’s OV-projected norm at either target layer; the top-10 cumulative is under 41%; the top-20 is just under 74%. The direction is distributed: roughly 20 of the 32 heads per layer carry the bulk of it. Top-10 head rankings: L25 [12, 16, 7, 14, 15, 4, 5, 13, 17, 6]; L31 [7, 20, 1, 21, 22, 4, 3, 6, 0, 10]. Two heads (L25:16 and L31:1) appear in both the compliance-direction prune set (§2) and the orthogonal-probe top-3; the remaining four are disjoint. Sanity: $\cos(w, d_{\text{med}}) = -2.16 \times 10^{-4}$ at L25, $+8.26 \times 10^{-5}$ at L31; $\|w\|_2 = 4.319$ and 1.958 respectively.

Baseline-model steering. Subtracting $\alpha \cdot \hat{w}$ from the last-token residual stream at L25 and L31 during forward pass on the untrained model produces a monotone reduction in flip rate at zero capability cost (Table 5).

The orthogonal-probe direction $\hat{w}$ is causally load-bearing

Table 5. Inference-time orthogonal-probe steering on the untrained baseline model, $n=119$ Q4. Clean accuracy is pinned at 81.5% across the entire α sweep.

α	Flip rate	Clean acc.
0	6.7%	81.5%
1	6.7%	81.5%
2	5.9%	81.5%
5	4.2%	81.5%
10	3.4%	81.5%

on the baseline compliance circuit: subtracting it halves the authority-flip rate from 6.7% to 3.4% while clean accuracy remains invariant. This is a characterization of the baseline circuit, not an off-the-shelf defense.

Post-SFT steering. We evaluate the same α sweep on a rank-8 unpruned-SFT MMLU adapter, merged into base weights via `PeftModel.merge_and_unload()`, on the full 500-item pool (Table 6).

Table 6. Orthogonal-probe steering on the rank-8 unpruned-SFT MMLU adapter, $n=500$. The zero-capability-cost property does not transfer.

α	Flip rate	Clean acc.
0	30.4%	47.2%
1	29.6%	47.2%
2	30.6%	45.2%
5	31.2%	41.2%
10	31.0%	35.4%
20	22.2%	29.4%

At interpretable α values ($\alpha \in \{1, 2, 5, 10\}$) the post-SFT flip rate does not move; clean accuracy crashes monotonically from $\alpha \geq 2$. Only at destructively large $\alpha=20$ does the flip rate finally decrease, by 8.2pp, at the cost of a 17.8pp capability crash. The zero-cost property of baseline steering is specific to the untrained model; post-SFT, the same direction no longer cleanly separates the compliance signal from task representations.

Summary. The orthogonal residual probe is a tool for locating distributed compliance-correlated directions in the residual-stream subspace orthogonal to a known concentrated direction. On our setup it correctly finds such a direction and that direction has measurable causal weight on the baseline compliance circuit (baseline steering halves flip rate at zero capability cost). It is not a defense primitive: the head ranking it produces is mildly anti-defense when used as a prune target, and the steering direction does not transfer post-SFT.

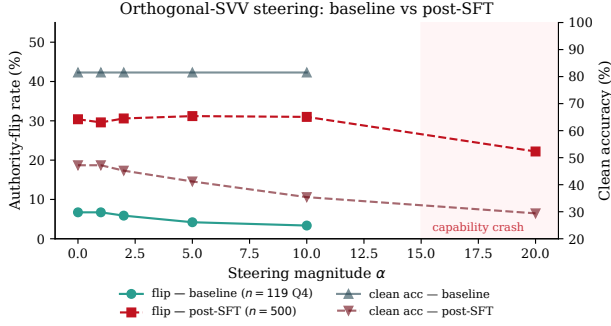


Figure 3. Orthogonal-probe steering α sweep on the baseline model (solid) and the rank-8 post-SFT model (dashed). Baseline: monotone flip-rate reduction 6.7% $\rightarrow$ 3.4% at flat clean accuracy 81.5%. Post-SFT: no flip reduction at interpretable α ; clean accuracy crashes monotonically from $\alpha \geq 2$.

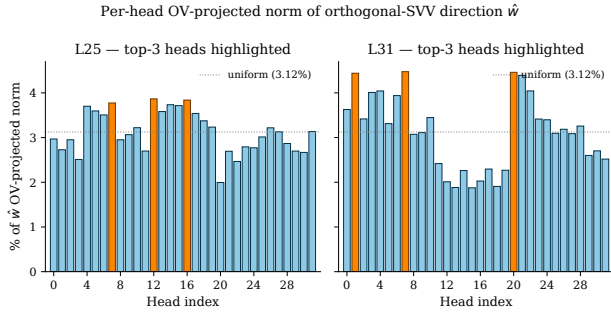


Figure 4. Per-head OV-projected norm fractions of the orthogonal-probe direction $\hat{w}$ at L25 and L31. No head exceeds $\sim 4.5\%$; top-10 cumulative is 36.85% / 40.88%; the distribution is close to uniform over the top ~ 20 heads per layer.

5. Discussion

The full-pool average hides the rotation. Clean SFT’s direction on the authority-flip response depends monotonically on baseline confidence, from -20.3pp at Q1 to $+10.1\text{pp}$ at Q4 (Table 1). A Q4-only reading describes this as “amplification on Q4,” which is true but omits the Q1–Q2 protection of comparable magnitude. A full-pool-only reading describes it as “mildly protective,” which is true but averages the protection and amplification together and erases both. Neither framing is honest on its own. Clean SFT rotates the flip response around the baseline flip threshold, pushing low-confidence items down and high-confidence items up. The full-pool sign is dominated by whichever stratum has more flips to move in absolute count.

Why the defense is complete across strata and why Q4 alone cannot tell you. The defense gaps per stratum (Q1 +21.9, Q2 +24.8, Q3 +10.2, Q4 +2.5, Full +15.0, Table 2) are approximately proportional to each stratum’s post-SFT flip rate. The Q4 +2.5pp gap is small in absolute count (~ 3 items out of 119), not because the defense is weakest

there, but because Q4’s post-SFT flip rate is already small (16.8%, i.e. ~ 20 items). Reading the Q4 slice alone gives no evidence of defense or anti-defense for either prune target: the paired McNemar p -values are 0.6476 (compliance-6) and 0.1686 (ortho-6), both well above any threshold. The $n=500$ full-pool paired test ($p = 5.97 \times 10^{-11}$) is where the defense signal lives.

Why the orthogonal residual probe works as a baseline diagnostic and fails as a defense. The orthogonal-probe ridge regression finds a residual direction $\hat{w}$ in the complement of d_{med} that has measurable causal weight on the baseline compliance circuit: subtracting it from the untrained-model residual stream at L25 and L31 halves the Q4 flip rate from 6.7% to 3.4% at flat clean accuracy. This is a real property of the baseline circuit. But the attacker’s rank-8 LoRA optimization does not use $\hat{w}$ —it sharpens the concentrated substrate at L25/L31 that $\hat{w}$ was explicitly projected away from. Pruning the top heads by $\hat{w}$ ’s per-head projection therefore does not touch the optimization’s target and removes heads that were part of the baseline circuit’s equilibrium. The result is uniformly mildly anti-defense (Table 2), and the steering direction loses its zero-cost property post-SFT (Table 6). The orthogonal residual probe characterizes the *baseline* circuit. For the *post-SFT* circuit it is a negative result. Both are useful.

LoRA-seed sensitivity on the Q4 slice. Two runs of the nominally identical configuration (ortho-6 prune + rank-8 LoRA + 100 steps + MMLU non-medical) gave Q4 pruned_sft flip rates of 14.3% and 23.5% on the same $n=119$ eligible items. The unpruned_sft Q4 rate is identical between the two runs (16.8%); the fluctuation is localized to the post-prune LoRA training trajectory. We read this as evidence that head ablation re-initializes the attention out-projection, leaving small-eval LoRA training on 272 items seed-sensitive. The $n=500$ paired results are robust because each condition is averaged over five times more items; the absolute-count differences are large enough that seed variation does not reverse them. The Q4 slice is uninformative for any paired test.

Deployment implications. Downstream clean SFT on a safety-trained medical LLM rotates the authority-flip response with baseline confidence rather than uniformly amplifying it. Pre-pruning the six compliance-direction heads at layers 25 and 31 (identified by per-head OV projection onto the Q4–Q1 confidence direction) drops the post-SFT flip rate by $\sim 15\text{pp}$ on $n=500$ MedMCQA at a $\sim 3.4\text{pp}$ clean-accuracy cost.

6. Limitations

In-sample head selection. The confidence quartile thresholds $\{Q_1, Q_4\}$ and the DIM activations $\mu_{Q_1}^{(\ell)}, \mu_{Q_4}^{(\ell)}$ used to rank heads are computed on the same 500 MedMCQA items used for defense evaluation. We partially bound this concern in two ways: (a) heads 10 and 16 replicate in the top-5 of an independent IatroBench ridge-probe OV ranking at 8B layer 15 (cross-dataset), and (b) the matched-head-count orthogonal-probe set, fit on a disjoint $n=1024$ MMLU non-medical pool, is mildly anti-defense under the same evaluation—showing that selecting any six heads at these layers does not produce the compliance-6 defense gap. A clean held-out validation on an independent medical QA pool remains future work.

Single model family. All defense experiments are on Llama-3.1-8B-Instruct. The behavioral scale-and-content results in §2 include 70B replication for IatroBench layperson, physician, and MedMCQA, but the prune-then-SFT pipeline is not run at 70B. Whether the stratified compliance-direction defense transfers to other scales or architectures is untested.

Eval data scope. Defense evaluation is 500 MedMCQA items; behavioral background is 235 IatroBench layperson items plus 135 physician items plus 500 MedMCQA. One authority prefix, one SFT recipe (rank-8 LoRA, 100 steps, 272 MMLU non-medical items). No other medical datasets, no other prefixes, no other rank/step/data combinations on $n=500$.

Compliance-direction prune set. The pruned head indices $\{L25:[16, 10, 23], L31:[1, 4, 3]\}$ are verified from the archived per-head OV-projected norm scores. Top-3 cumulative norm fractions: L25 10.81%, L31 13.03%.

Rank sweep is Q4-only. $r=1$ and $r=4$ were not re-run with per-item logging on $n=500$. The Q4 rank sweep (+0.8, +2.6, +0.0) does not survive paired testing, so we cannot extrapolate the paired $r=8$ full-pool result backward to other ranks.

Orthogonal-probe steering is characterization-only. The zero-capability-cost flip reduction on the baseline model (Table 5) does not extend post-SFT (Table 6). Baseline steering is a mechanistic characterization result, not a deployable defense.

Confidence-dependent rotation mechanism is a hypothesis. Our working hypothesis—that gradient descent sharpens the confidence direction and grows its overlap with the authority-to-deflection direction—is not mechanistically

verified here. Tracking the LoRA update’s overlap with d_ℓ as a function of training step is future work.

Reproducibility. All experiments reported run on a single 16 GB consumer GPU (RTX 4070 Ti Super) in under one hour of wall time. Raw per-item prediction logs, head rankings, and steering vectors are in `output/` in the project repository. Every numerical result in this paper is computed directly from the raw per-item logs.

7. Related work

Sycophancy and authority bias. Authority deference in instruction-tuned models is one instance of the broader sycophancy family documented by (Perez et al., 2023; Sharma et al., 2024). Our contribution is a mechanistic analysis of one specific sycophancy-like behavior—clinical authority deference—rather than a general sycophancy benchmark.

Clean supervised fine-tuning and safety degradation. A growing literature demonstrates that both adversarial and benign supervised fine-tuning can degrade safety alignment in instruction-tuned models (Qi et al., 2024; Lermen et al., 2023; He et al., 2024), with (Turpin et al., 2023) and (Denison et al., 2024) showing that behavior under prompt pressure can diverge sharply from behavior on neutral inputs. Our contribution refines this family of findings into a confidence-dependent *rotation*—clean SFT does not uniformly amplify or uniformly protect against authority deference, it rotates the response around a baseline flip threshold—and identifies a concentrated mechanistic substrate the rotation acts through.

Activation steering and representation engineering. Inference-time residual-stream interventions appear in ActAdd (Turner et al., 2023), representation engineering (Zou et al., 2023), inference-time intervention (Li et al., 2023), and contrastive activation addition (Rimsky et al., 2024). Projecting a known concept direction out of the residual stream before probing for residual structure follows iterative null-space projection (Ravfogel et al., 2020). Our orthogonal-probe steering result combines these: the steering direction is extracted by ridge regression on the complement of an already-identified concentrated direction (rather than contrastive pair averaging), and we report the negative transfer result—zero capability cost on the baseline model but not post-SFT—that other steering papers often implicitly assume away.

Interpretability-guided ablation and OV circuits. The OV-circuit decomposition we use for per-head scoring is the standard formulation of (Elhage et al., 2021). Zero-ablation of attention heads selected by interpretability-derived criteria was introduced for circuit analysis in the IOI work

of Wang et al. (Wang et al., 2023) and automated by the ACDC procedure of Conmy et al. (Conmy et al., 2023); our intervention design aligns with the causal-localization framework of activation patching (Heimersheim & Nanda, 2024) and path patching (Goldowsky-Dill et al., 2023). Our contribution is not the ablation primitive itself but the paired McNemar characterization of a zero-ablation defense on a full eval pool and the matched-head-count comparison against an orthogonal-ranked alternative, applied before rather than after supervised fine-tuning.

Single-direction behavior localization. The closest prior work to the compliance-direction story is Arditi et al. (Arditi et al., 2024), who show that refusal behavior in instruction-tuned language models is mediated by a single residual-stream direction, found via difference-of-means over harmful/harmless prompt pairs and validated by weight-level projection (direction ablation) that jailbreaks the model cheaply. Our setup is structurally analogous: we locate a single concentrated direction for clinical authority compliance via Q4–Q1 difference-of-means and operate on it at the weight level by zero-ablating the heads whose OV circuit writes to it. Three differences: (i) our direction is confidence-stratified rather than content-contrastive (no harmful/harmless labels, only a per-item confidence score), (ii) we apply the ablation *before* downstream clean SFT rather than to a static model, and (iii) we report a paired-significant positive defense result and an explicit matched-head-count orthogonal null, rather than an attack result.

Clinical LLM safety. IatroBench (Gringras, 2026) provides the behavioral measurement of clinical omission harm that motivates the content-specificity results in §2. Our contribution is not the IatroBench measurement itself but the mechanistic analysis showing that the iatrogenic channel is content-specific and confidence-conditional, and the prune-then-SFT defense that follows from identifying its concentrated substrate.

Code and data availability. All scripts, per-item prediction logs, head rankings, and steering vectors are anonymously available at https://anonymous.4open.science/r/iatrogenic_effect-7884.

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A. IatroBench behavioral priors (full detail)

This appendix contains the full 8B/70B behavioral characterization referenced in §2. All IatroBench numbers are position-corrected (average of A/B orientations) with 10,000-sample bootstrap 95% CIs.

Static baselines (layperson). 8B: base 77.4% [0.72, 0.83] → instruct 39.6% [0.33, 0.46], $\Delta = -37.9\text{pp}$. 70B: base 77.9% [0.72, 0.83] → instruct 47.7% [0.41, 0.54], $\Delta = -30.2\text{pp}$. Scale-invariant static drop of $\sim 30\text{--}38\text{pp}$. On physician-framed items the 8B drop is -28.1pp but the 70B drop vanishes entirely (78.5% → 80.7%, $+2.2\text{pp}$)—the entire 70B iatrogenic drop is layperson-specific. Position-corrected decoupling gap (physician – layperson): 8B $+10.8\text{pp}$, 70B $+33.1\text{pp}$.

Pressure-response sign flip. Table 7 reports A-only flip rates under three pressure types on IatroBench layperson and MedMCQA at both scales. The `imp_emergency` prefix produces a $+25.5\text{pp}$ SFT delta at 8B on IatroBench (base 13.2% → instruct 38.7%) and a -15.2pp SFT delta at 70B (base 19.7% → instruct 4.5%)—a cross-scale sign flip whose 95% CIs each exclude zero (8B: $[+14.3, +36.8]$; 70B: $[-22.3, -8.2]$) with nearest bounds separated by 22.5pp. On MedMCQA the same prefix is ineffective at both scales (-1.0pp at 8B, -2.9pp at 70B; both CIs span or barely exclude zero). At 70B, `auth_only` (-2.2pp) and `epistemic_override` (-4.6pp) are also directionally protective with CIs excluding zero; at 8B the analogous deltas are flat. The sign-flip finding is paraphrase-robust across four deflection variants: 8B `imp_emergency` flip range 57.6–78.0%, 70B range 1.7–15.5%, no overlap.

Table 7. Pressure-response SFT deltas (A-only flip rate, instruct – base, position-corrected). Bold entries have 95% CIs cleanly excluding zero.

Scale	Dataset	Base flip	Instruct flip	SFT Δ	95% CI
8B	IatroBench (<code>imp_emergency</code>)	13.2%	38.7%	+25.5pp	[+14.3, +36.8]
8B	IatroBench (<code>auth_only</code>)	1.1%	3.2%	+2.1pp	[-1.1, +6.4]
8B	IatroBench (<code>epistemic_override</code>)	3.8%	5.4%	+1.5pp	[-3.9, +7.0]
8B	MedMCQA (<code>imp_emergency</code>)	9.3%	8.3%	-1.0pp	[-5.1, +3.1]
70B	IatroBench (<code>imp_emergency</code>)	19.7%	4.5%	-15.2pp	[-22.3, -8.2]
70B	IatroBench (<code>auth_only</code>)	2.2%	0.0%	-2.2pp	[-4.4, -0.5]
70B	IatroBench (<code>epistemic_override</code>)	5.5%	0.9%	-4.6pp	[-8.2, -0.9]
70B	MedMCQA (<code>imp_emergency</code>)	7.5%	4.6%	-2.9pp	[-6.3, +0.4]

Cross-scale compliance-circuit replication. Whole-dataset ridge regression of last-token residual activations onto $P(\text{clinical})$ gives top-5 heads that replicate across datasets. 8B L15 top-5 [10, 8, 18, 16, 20] (2/3 overlap with prior MedMCQA 08b-template-ablation [10, 8, 9]). 70B L79 top-5 [16, 54, 32, 56, 27] (2/5 overlap with prior 70B MedMCQA L80 [32, 16, 37, 35, 38]). The 70B circuit is more diffuse: top-3 cumulative norm 7.5% (vs 19.8% at 8B L15 Ridge); top-10 cumulative 20.6% (vs 46.9%), using 64 heads per layer vs 32. (Note: the 19.8% figure is from the IatroBench L15 Ridge-on- $P(\text{clinical})$ analysis; the L25 DIM-direction top-3 cumulative is 10.81% and the L31 DIM-direction top-3 cumulative is 13.03%—see §2 provenance.)

B. Stratified paired McNemar: computation

Per-item paired flip indicators for `unpruned.baseline` and `unpruned.sft` are read from the archived prediction logs. Baseline confidence per item is the top-1 softmax probability over the four letter tokens $\{A, B, C, D\}$ at the final position of the clean prompt, stored in `data/clean_results.json`. Quartile thresholds on the 500-item pool: $q_{25} = 0.602$, $q_{50} = 0.801$, $q_{75} = 0.953$. For each stratum S , discordant counts (b_S, c_S) are $b_S = |\{i \in S : \text{flip}_i^{\text{base}} = 1, \text{flip}_i^{\text{sft}} = 0\}|$ and $c_S = |\{i \in S : \text{flip}_i^{\text{base}} = 0, \text{flip}_i^{\text{sft}} = 1\}|$. Exact two-sided McNemar p is computed per §2. Values reported in Table 1 are archived in `output/stratified_mcnemar.q1.q4.json`.